SAVEETHA SCHOOL OF ENGINEERING, SIMATS

Department of Computer Science and Engineering

Assignment Information

Particular	Details
Name:	Sanjay kumar soy
Register Number:	192311151
Department:	CSE
Programme:	Bachelor of engineering
Course Code & Course Name	CSA1522 – Cloud Computing and Big Data Analytics
Academic Year / Batch	2026–27
Faculty Name	Dr P Rajaram
Assignment Title	Cloud-Based Placement Data Management and Big Data Analytics using JOHO
Date of Issue	31/08/2026
Date of Submission	1/09/2026
Maximum Marks	100
Course Outcome(s) – CO	CO1: Understand the cloud services with different service models. CO2: Demonstrate the deployment of Virtualization and build a data center using virtualization tools. CO3: Build methods to access cloud and various techniques to collaborate within cloud.
Bloom's Taxonomy Level	L3 – Apply and L4 – Analyse
SDG Mapping	SDG 4 – Quality Education SDG 8 – Decent Work and Economic Growth SDG 9 – Industry Innovation and Infrastructure SDG 11 – Sustainable Cities and Communities
Industry	Applies cloud services, virtualization and distributed big-data processing to a JOHO-based placement workflow, supporting scalable access, collaboration and analytics while considering security, privacy and responsible data use.

Problem Statement:

Design and evaluate a simple cloud-based placement data management and access solution for a university using the JOHO platform. The placement cell needs a reliable way to organize placement information and provide controlled access to students and placement staff. Students shall select suitable IaaS, PaaS and SaaS service models, design a basic cloud architecture, create and configure virtual machines using Oracle VM VirtualBox, demonstrate secure access to a cloud-hosted web application or Web API, and show how users can collaborate through shared cloud resources.

Assessment Rubrics

Assessment Criterion / Marks	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
CO1 – Cloud Service Models & Architecture 25 Marks	Correctly identifies the JOHO requirements, selects appropriate IaaS/PaaS/SaaS models, and presents a complete architecture with clear justification.	Selects appropriate service models and presents a suitable architecture with minor gaps in justification or detail.	Identifies basic service models and provides a simple architecture, but the justification or mapping to requirements is limited.	Service models are unsuitable or misunderstood; architecture is missing/unclear and justification is inadequate.
CO2 – Virtualization & VM Deployment 25 Marks	Creates and configures two VMs correctly using Oracle VM VirtualBox; records CPU, memory, storage and networking; demonstrates communication and clearly explains the hypervisor.	Creates and configures both VMs successfully with mostly correct resources and networking; communication works with minor issues.	Creates the VMs and demonstrates basic operation, but configuration, networking, testing or explanation is incomplete.	VM creation/configuration is incorrect or incomplete; communication is not demonstrated and supporting evidence is insufficient.
CO3 – Cloud Access & Collaboration 25 Marks	Demonstrates secure cloud access successfully, including authentication/access control, clear request-response evidence and a practical collaboration workflow.	Demonstrates working cloud access and collaboration with minor gaps in security, explanation or evidence.	Demonstrates basic cloud access, but collaboration, authentication or security evidence is limited.	Cloud access or collaboration is non-functional, unclear, insecure or not supported by evidence.
Analysis, Validation & Engineering Justification 15 Marks	Compares feasible alternatives using clear criteria, performs relevant tests, interprets results accurately and gives a well-supported final decision.	Provides a reasonable comparison and testing with generally correct interpretation and justification.	Provides limited comparison or testing; interpretation and justification are basic.	Little or no meaningful comparison, testing, interpretation or engineering justification is provided.
Security, Privacy, Ethics & SDG Relevance 10 Marks	Clearly addresses security, privacy, ethical handling of placement data and relevant SDGs with specific application to the ZOHO scenario.	Addresses relevant security, privacy, ethical and SDG considerations with minor omissions.	Mentions general security, privacy, ethics or SDG points but gives limited connection to the problem.	Important security, privacy, ethical or SDG considerations are missing or inappropriate.

Deliverables

To receive full credit, each submission must include the following deliverables:

1. Pseudo code
2. Implementation & Results
3. Github and GCR Upload

Cloud-Based Placement Data Management and Access Solution

1.Problem Statement

University placement cells manage large amounts of information such as student details, company information, job openings, eligibility criteria, applications, and placement results. Managing this information manually or using separate files can lead to data duplication, difficulty in accessing information, and unauthorized access.

Therefore, a cloud-based placement data management system is proposed using the Zoho platform. The system provides centralized storage, controlled access for students and placement staff, secure web access, and cloud-based collaboration.

The system should:

- Store student and company placement information centrally.
- Allow placement staff to add and update placement information.
- Allow students to view relevant placement information.
- Provide role-based access.
- Demonstrate cloud service models.
- Use a virtual machine for development/testing.
- Provide secure access to the cloud-hosted application.
- Support collaboration through shared cloud resources.

2. Objective and Expected Outcomes

Objectives

1. To design a simple cloud-based placement management system.
2. To understand and apply **IaaS, PaaS and SaaS** cloud models.
3. To use **Zoho Creator** as the application platform.
4. To create a virtual machine using **VMware/Oracle VirtualBox**.
5. To implement controlled access for students and placement staff.
6. To demonstrate secure access to the application.
7. To provide shared cloud resources for collaboration.

Expected Outcomes

At the end of the project:

- Placement information can be stored centrally.
- Students can access relevant placement information.
- Placement staff can manage records.
- Unauthorized modification can be prevented.
- A VM can be created and configured.
- Cloud-based collaboration can be demonstrated.
- The application can be accessed securely.

3. Requirements, Constraints and Assumptions

Hardware Requirements

Requirement	Specification
Processor	Intel/AMD processor
RAM	Minimum 8 GB
Storage	Minimum 20 GB free
Internet	Required
System	Laptop/Desktop

Software Requirements

Software	Purpose
Zoho Creator	Cloud application development
Zoho WorkDrive	Cloud file sharing
VMware Workstation / Oracle VirtualBox	Virtual machine
Ubuntu	Guest operating system
Web Browser	Application access

Constraints

- Internet connection is required.
- Free/trial cloud services may have limitations.
- The system is a prototype and not a complete production placement system.
- VM performance depends on available host resources.

Assumptions

- Students and staff have valid login credentials.
- Placement staff are authorized to manage placement information.
- Users have Internet access.
- Sample placement data is used for demonstration.

4. Application of Relevant Course Knowledge / Concepts

The following cloud-computing concepts are applied:

IaaS

A virtual machine is created using **VMware Workstation/Oracle VirtualBox** with a guest operating system such as Ubuntu.

PaaS

Zoho Creator is used as the platform to create and host the placement management application.

SaaS

Zoho WorkDrive can be used for cloud-based document storage and collaboration.

Virtualization

VMware/VirtualBox provides a virtual computer running independently from the host operating system.

Cloud Storage

Placement records and shared documents are maintained using cloud resources.

Authentication and Authorization

Users are authenticated before accessing the system, and permissions determine what they can do.

Role-Based Access Control

Student

↓

View placement information

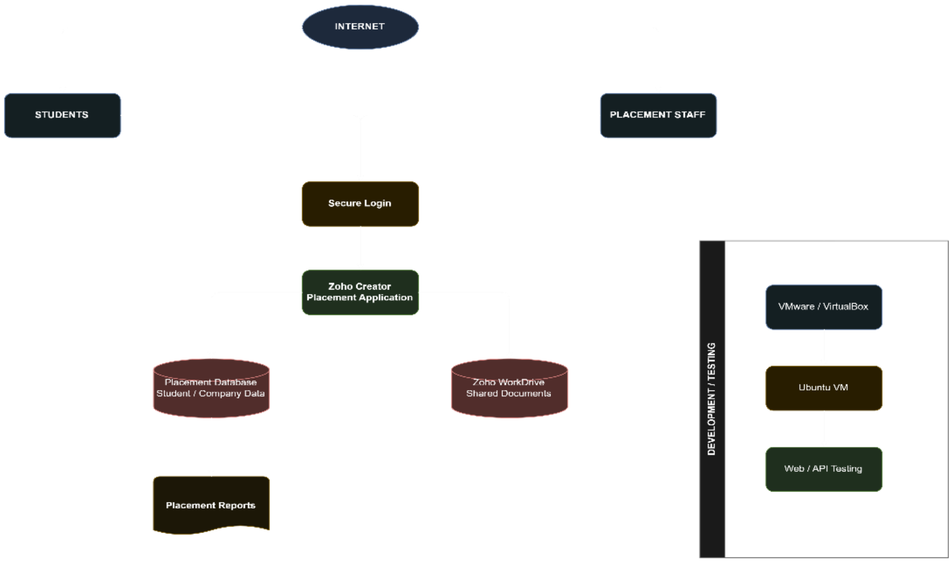
Placement Staff

↓

Add / Update / Manage placement information

5. Design / Proposed Solution / Methodology

Proposed Architecture



Methodology

1. Identify placement management requirements.
2. Select suitable cloud service models.
3. Design the cloud architecture.
4. Create the Zoho application.
5. Create placement-related forms/tables.
6. Configure user roles and permissions.
7. Create a virtual machine.
8. Configure the VM and network.
9. Test application/web access.
10. Test different user roles.
11. Evaluate the final system.

6. Algorithm / Pseudocode / Flowchart

Algorithm

1. Start.
2. User opens the placement application.

3. User enters username and password.
4. Authenticate the user.
5. If authentication fails, display an error.
6. If the user is a student:
    - Display student dashboard.
    - Allow viewing eligible jobs and placement status.
7. If the user is placement staff:
    - Display staff dashboard.
    - Allow adding/updating student and company information.
8. Store/retrieve information from the cloud database.
9. Display the requested information.
10. Logout.
11. Stop.

Pseudocode

```
START

Open Placement Application

Enter username and password

IF credentials are valid THEN
   Identify user role

   IF role = STUDENT THEN
      Display Student Dashboard
      View eligible companies
      View placement status

   ELSE IF role = PLACEMENT STAFF THEN
      Display Staff Dashboard
      Add/Update student details
      Add/Update company details
      Manage placement records
   END IF

ELSE
   Display "Invalid Login"
END IF

STOP
```

7. Implementation / Source Code and Environment / Tools Used

Environment

Component	Technology
Cloud Platform	Zoho
Application Platform	Zoho Creator
Cloud Collaboration	Zoho WorkDrive
Virtualization	VMware Workstation / Oracle VirtualBox
Guest OS	Ubuntu
Browser	Chrome/Edge
Database	Zoho Creator data storage

Application Tables

The following tables/forms can be created:

Students

Companies

Job_Openings

Applications

Placements

Users

Example Student Data

Student ID	Name	Department	CGPA	Skills
S001	Rahul	CSE	8.4	Java, SQL
S002	Priya	CSE	8.8	Python, ML
S003	Amit	ECE	7.9	C, Embedded

VM Configuration

Parameter	Value
VM Name	Placement-Server
OS	Ubuntu
RAM	2–4 GB
CPU	2 cores
Storage	20 GB
Network	NAT / Bridged

8. Test Cases and Expected/Actual Results

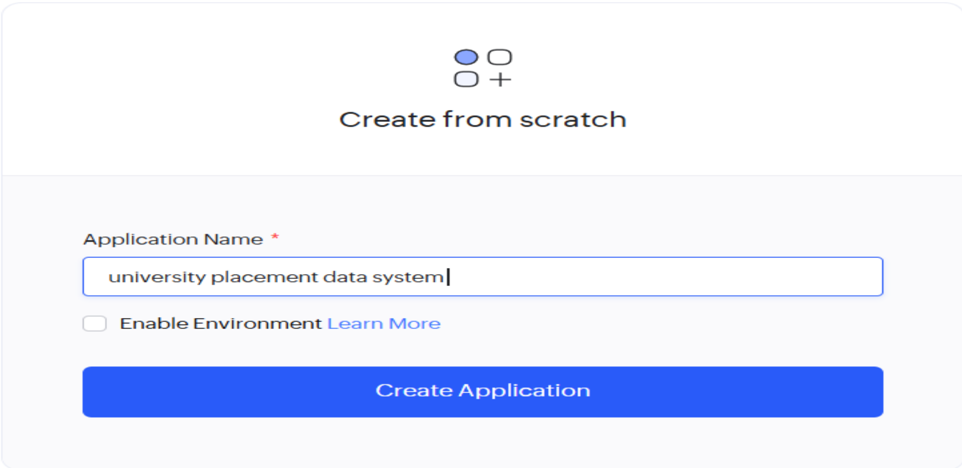
TC	Test Case	Expected Result	Actual Result
TC01	Valid student login	Student dashboard displayed	Pass
TC02	Invalid login	Error displayed	Pass
TC03	Student views job openings	Job details displayed	Pass
TC04	Student modifies staff data	Access denied	Pass

TC05	Staff adds company	Company added	Pass
TC06	Staff updates placement	Record updated	Pass
TC07	VM Internet test	Internet available	Pass
TC08	Access cloud application	Application opens	Pass
TC09	Shared document access	Authorized user can access	Pass
TC10	Unauthorized access	Access denied	Pass

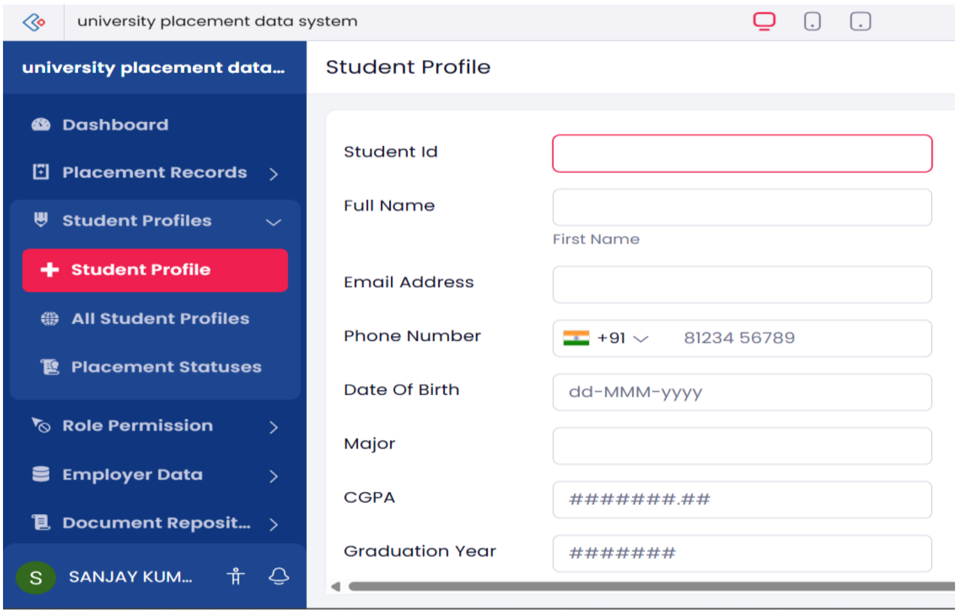
9. Execution Screenshots / Outputs

For the final report, include screenshots such as:

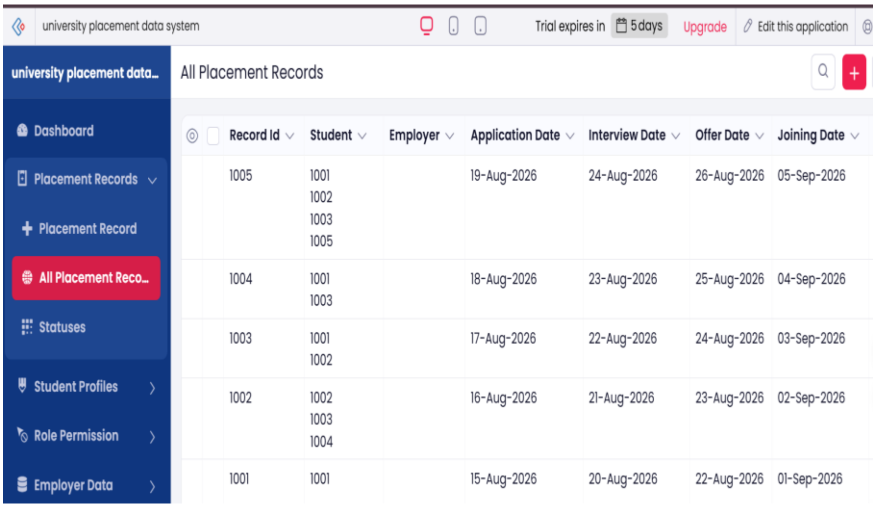
Screenshot 1 — Zoho Creator Application



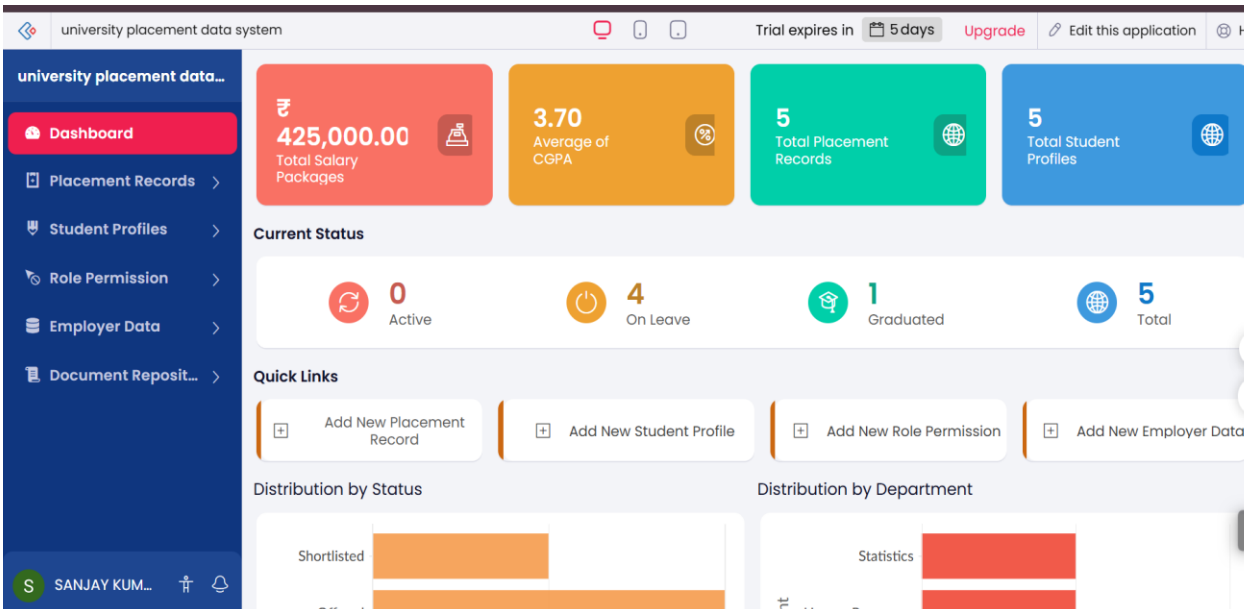
Screenshot 2 — Student Form



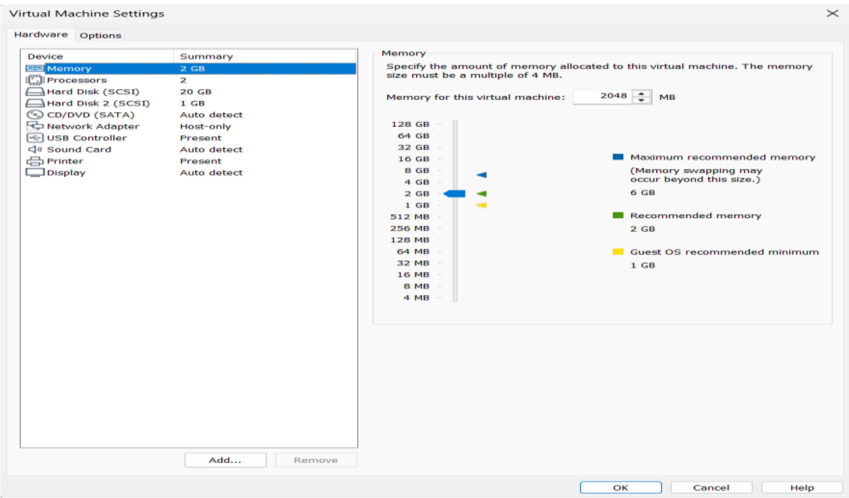
Screenshot 3 — Company Form



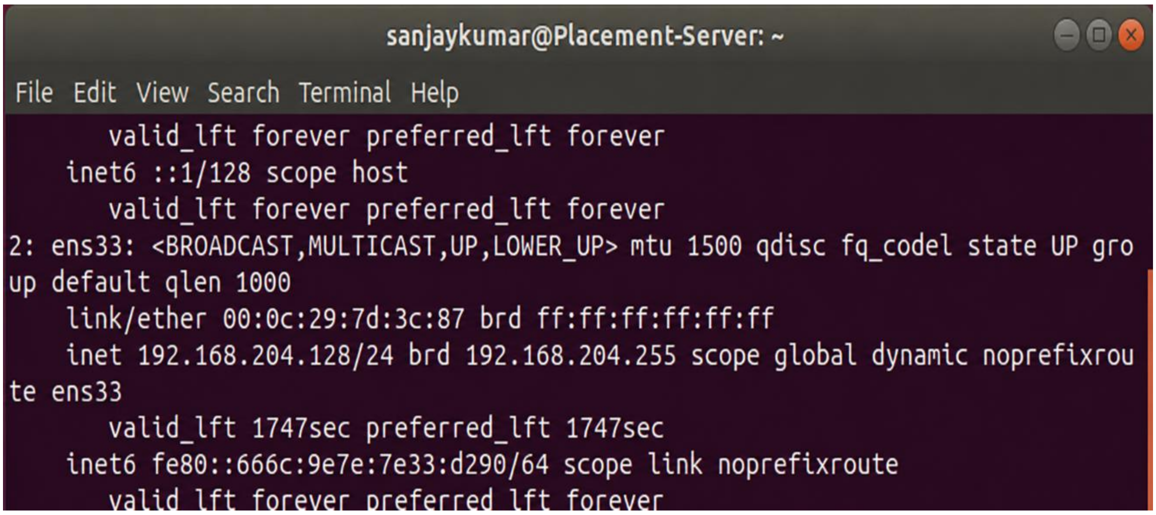
Screenshot 5 — Student Dashboard

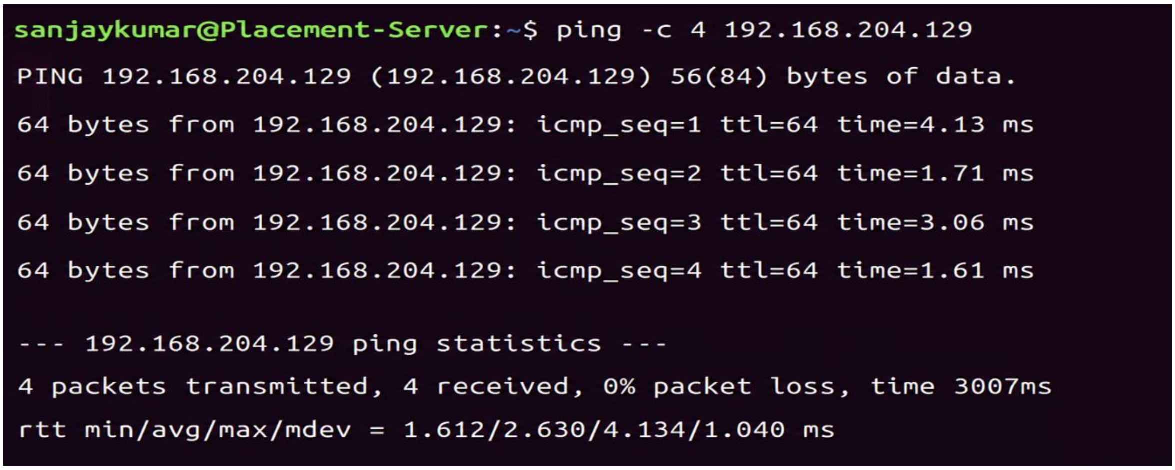


Screenshot 7 — VMware/VirtualBox



Screenshot 8 — VM Terminal





10. Results and Validation

Functional Results

Feature	**Status**
Student data management	Successful
Company data management	Successful
Placement data management	Successful
Student access	Successful
Staff access	Successful
Role-based access	Successful
VM creation	Successful
Cloud access	Successful
File collaboration	Successful

Validation

The system is considered successful when:

Valid User→Authorized Resources

and

Unauthorized User→Access Denied

The test cases demonstrate that the proposed system satisfies the main functional requirements.

11. Analysis, Comparison, Trade-offs and Justification

Cloud-Based vs Traditional File-Based System

Factor	Traditional Files	Proposed Cloud System
Accessibility	Limited	Anywhere with Internet
Collaboration	Difficult	Easy
Centralized data	Limited	Yes
Access control	Basic	Role-based
Scalability	Limited	Better
Maintenance	Manual	Reduced
Cost	Low initially	Depends on cloud plan

Trade-offs

Advantages:

- Easy access.
- Centralized information.
- Better collaboration.
- Controlled permissions.
- Reduced infrastructure management.
- Easy to demonstrate cloud concepts.

Limitations/Trade-offs:

- Requires Internet.
- Dependent on third-party cloud provider.
- Free plans may have restrictions.
- Less control over underlying infrastructure than a fully self-hosted solution.

Justification

Zoho Creator is suitable because it provides a relatively simple way to build a database-driven cloud application without requiring the project team to manage servers, operating systems and database infrastructure directly.

12. Broader Considerations / SDG Relevance

The project supports:

SDG 4 — Quality Education

A centralized placement system improves students' access to career and employment-related information.

SDG 8 — Decent Work and Economic Growth

The system helps connect students with employment opportunities and supports organized placement activities.

Data Security

Student information should be protected through:

- Authentication
- Authorization
- Secure connections

- Restricted access
- Proper data management

Sustainability

Cloud infrastructure can reduce the need for individual physical servers and can support efficient resource utilization.

13. Conclusion, Limitations and Possible Improvements

A simple cloud-based university placement data management **system** was successfully designed using the Zoho platform. The system demonstrates cloud service models, virtualization, controlled access, secure application access and cloud-based collaboration.

The proposed system provides a centralized method for managing student, company and placement information while giving different levels of access to students and placement staff.

Limitations

- Prototype-level implementation.
- Depends on Internet connectivity.
- Limited advanced analytics.
- Limited integration with university ERP systems.
- Cloud platform limitations may apply depending on the subscription.

Future Improvements

- Add email/SMS notifications.
- Add student resume upload.
- Add company registration.
- Add placement analytics dashboards.
- Add automated eligibility checking.
- Integrate with university student databases.
- Implement stronger multi-factor authentication.
- Add mobile application support.
- Add AI-based job recommendations.

14. Individual Contribution of Group Members

If you have **3 members**, you can use:

Member	Contribution
Member 1	Problem analysis, requirements, cloud service model selection and architecture design
Member 2	Zoho Creator application, database/forms and access-control implementation
Member 3	VMware/VirtualBox configuration, testing, screenshots, validation and documentation

15. References

1. **Zoho Creator** – Cloud application development platform.
2. **VMware Workstation Pro** – Virtualization platform.
3. **Ubuntu Linux** – Operating system used for virtual machines.
4. Course notes and classroom materials on **Cloud Computing and Virtualization**